2.1.3

Title: How do disparate urbanization and climate change imprint on urban thermal variations? A comparison between two dynamic cities in Southeast Asia [PAYWALLED] [ABSTRACT ONLY]
Author: Can Trong Nguyen, Amnat Chidthaisong , Atsamon Limsakul, Pariwate Varnakovida, Chaiwat Ekkawatpanit Phan Kieu Diem , Nguyen Thi Hong Diep

Abstract: This study assessed the combined influences of climate change and urbanization on land surface temperature (LST) and surface urban heat island (SUHI) in the Bangkok metropolitan region (BKK) and Ho Chi Minh City metropolitan area (HCM) during 1990–2020. We found climate change evidence from trends and variations of temperature extremes in both cities throughout the past three decades, especially for the significant upward trends of nighttime temperature in HCM (0.23 – 0.3 °C/decade). These cities have been undergoing rapid urbanization, in which HCM has a higher urbanization rate with an annual growth rate of 8.03%/year against BKK (4.26%/year). Landscape metrics have rapidly shifted towards urban aggregation and green space fragmentation along with urbanization. A peer comparison shows that BKK holds a greater urban size and LST intensity, while HCM leads in urban growth and SUHI magnitude. These changes jointly contribute to climbing SUHI intensity, especially the dramatic influence of surface changes. Yet, temperature extremes also play an indispensable role in regulating SUHI magnitude, in which the impact in HCM is considerable compared to the effect in BKK. The research findings provide crucial information for both cities – where incorporation of climate change resilience strategies and SUHI mitigation into the master plans for livable cities are recommended.

Title: Thermal intensification and urban land cover change in Southeast Asian capitals: over three decades of satellite observations and trend analysis
Author: Yong Le Ng, Ming Han Lim, Yuk Feng Huang & Yee Ling Lee

On multi-decadal satellite trends and specific warming rates:
"Significant warming trends emerged, especially in Vientiane ($0.350^{\circ}C/year$), Kuala Lumpur ($0.235^{\circ}C/year$), and Bangkok ($0.215^{\circ}C/year$). LULC classification revealed major urban expansion with vegetation and water loss. This first multi-decadal, cross-city IUHII assessment in tropical Southeast Asia addresses a key research gap."

On rapid impervious expansion, vegetative loss, and thermal intensification in tropical/equatorial climates:

"Rapid urban expansion in the region's capital cities has been accompanied by extensive loss of vegetation, proliferation of impervious surfaces, and sustained anthropogenic heat release, all within a humid tropical climate highly sensitive to land cover perturbations."

"A steady upward trend in LST was detected across all land cover classes throughout the 34 years. Urban expansion at the expense of vegetated and aquatic areas has weakened natural cooling mechanisms, intensified heat storage, and altered surface energy balances. [...] Across all cities, urban surfaces

consistently exhibited the highest LST values, while water bodies remained the coolest, reinforcing the role of land cover composition in regulating surface thermal environments."

Title: Cooling the cities: A comprehensive review of urban heat island mitigation strategies in Southeast Asia
Author: Kyaw Zabu Tun, Malay Pramanik, Suraj Kumar Mallick, Rabin Chakrabortty, Bijay Halder, Kanak N. Moharir, Chaitanya Baliram Pande, Mohamed Zhran

"This systematic review synthesizes 27 peer-reviewed studies (2020–2023) on urban heat island (UHI) mitigation in Southeast Asia, focusing on cooling effectiveness and implementation feasibility. In dense cities like Singapore and Kuala Lumpur, green infrastructure—urban forests, green roofs, and street-tree networks—proves most effective, reducing air temperatures by up to 2.6 °C and surface temperatures by 11 °C. Engineered solutions, such as high-albedo coatings and permeable pavements, offer 4–9 °C of surface cooling and added benefits like stormwater control, though tropical humidity, glare, and maintenance pose challenges. Informal settlements demonstrate the value of low-cost, community-led cooling efforts using shade structures and vegetation. Despite these benefits, barriers persist: high costs, limited urban land, and fragmented governance systems hamper the scalability of hybrid green-grey solutions. Evaporation-based strategies are further constrained by complex urban geometries and monsoonal climates. Fewer than 20 % of Southeast Asian cities incorporate UHI mitigation into formal planning frameworks, limiting systemic uptake. To advance climate-resilient urbanism, interdisciplinary collaboration is needed. Future priorities should include long-term monitoring in tropical climates, integrated trials of hybrid interventions, and cost–benefit models that capture health, energy, and flood resilience co-benefits. Embedding these approaches into planning systems, as seen in Bangkok's zoning reforms and Singapore's Green Mark, is key to sustainable adaptation."

Title: The effect of urbanization on temperature indices in the Philippine
Author: John A. Manalo, Jun Matsumoto, Hiroshi G. Takahashi, Marcelino Q. Villafuerte II, Lyndon Mark P. Olaguera, Guoyu Ren, and Thelma A. Cinco
DOI: 10.1002/joc.7276

Abstract
"This paper presents a comprehensive analysis of the effect of urbanization on the surface air temperature (SAT) from 1951 to 2018 in the Philippines. The daily minimum temperature (Tmin) and daily maximum temperature (Tmax) records from 34 meteorological stations were used to derive extreme temperature indices. These stations were then classified as urban or rural based on satellite night-lights. The results showed a significant difference in the SAT trends between urban and rural stations, indicative of the effect of urbanization in the country. Larger and more significant warming trends were observed in indices related to Tmin than those related to Tmax . In particular, the effects of urbanization were significant in the annual index series of Tmin , diurnal temperature range, minimum Tmin , percentage of days when Tmin was less than the 10th percentile (TN10p), percentage of days when Tmin was greater than 90th percentile (TN90p), and the number of coldest nights. The effects of urbanization were not as clear on the

index series of maximum Tmax (TXx), minimum Tmax (TXn), percentage of days when Tmax was less than 10th percentile (TX10p) and the number of hottest days. The effects of urbanization on the annual series of extreme temperature indices were statistically significant at the 95% confidence level, with the exception of Tmax , TXn , TXx , TX10p , and the number of hottest days. Further analysis revealed that the effect of urbanization was the greatest during the DJF (December-January-February) season."

Trends in the Extreme Temperature Indices
"A 0.09∘C/decade difference in the annual Tmin was observed between urban and rural stations, indicating that the classification of the stations is able to illustrate the effects of urbanization in the long-term temperature trends in the Philippines. The results also show that the Tmin and Tmax trends for rural stations were both 0.16∘C/decade, while Tmin showed a larger increase compared to Tmax for urban stations, with a difference equivalent to 0.05∘C/decade. This implies that urbanization impacts night-time temperatures more than daytime temperatures..."

"The observed relatively higher trends in Tmin compared to those in Tmax contributed to a decreasing trend in the annual DTR series of approximately −0.04∘C/decade; this was particularly evident following the early 1990s (Figure 4d)."

Urbanization Effect and Contribution
"In terms of seasonal analysis, almost all extreme temperature indices were found to have a higher effect of urbanization during the DJF season, with the exception of Tave , TNn , TXx , TX90p , and TN90p . This might be related to a possibly lower and more stable urban boundary layer (UBL) due to lower temperature measurements observed during DJF as compared to other seasons (Villafuerte et al., 2020)."

Conclusions
"This study offers an in-depth analysis of the effect of urbanization on long-term trends of extreme temperature indices in the Philippines from 1951 to 2018. The results of this study were used to draw the following conclusions:"

"A higher trend in Tmin of approximately 0.25∘C/decade was observed in urban stations, while this was approximately 0.16∘C/decade in rural stations. This indicates a clear influence of the enhanced UHI effect on night-time temperatures detected at urban stations."

"Larger and more significant effects of urbanization were observed on indices related to Tmin , such as TNn , TNx , TN10p , TN90p , and coldest nights, compared to indices related to Tmax . Seasonal analysis has shown that the greatest effect of urbanization is observed during the DJF season."

Title: Modeling the influence of land cover dynamics on spatio-temporal variations in land surface temperature in Cagayan de Oro River basin, Mindanao, Philippines [PAYWALLED] [ABSTRACT]
Author: Rylle Adriane Galvez, Hilly Ann Roa-Quiaoit, Frandel Louis Dagoc, Jaime Guihawan & Peter Suson

“The onset of agricultural intensification and urban expansion in Cagayan de Oro

River basin has been continuously changing its landscape over the years. While it undergoes modifications its land cover, its associated impact on the land surface temperature over the years has not been evaluated. This study aims to model the spatio-temporal dynamics of land cover (2003–2020) and determine its influence on the land surface temperature variations in Cagayan de Oro River basin, Mindanao, Philippines from 2003 to 2020 using geographical information systems (GIS) and remote sensing techniques. Significant changes were observed in annual and perennial croplands which experienced substantial increases and is preceded by built-up areas. Considerable decreases were however recorded in shrubland, grassland, mangroves, and open forest. Between 2003 and 2020, the mean land surface temperature in Cagayan de Oro River basin increased by 5.48 °C. The results indicate that built-up areas and bareland consistently recorded the highest mean LST values, while water bodies, mangroves, and forest cover had the lowest temperatures. This study promotes evidence-based decision-making for sustainable land use planning and management to mitigate the increasing land surface temperature in Cagayan de Oro River basin.”

Title: Land-use/land cover change and land surface temperature in Metropolitan Manila, Philippines using landsat imagery [PAYWALLED] [ABSTRACT]
Author: King Joshua Almadrones-Reyes & Nikki Heherson A. Dagamac

“Remote sensing technology has been widely used to address the issue of urbanization for monitoring. In Metropolitan Manila, Philippines, there is a decline in vegetation over the last decade. This leads to the excess heat in the surface temperature and a phenomenon known as Urban Heat Island. In this study, the land surface temperature (LST), land-use/land cover changes (LULC), normalized difference vegetation index (NDVI) and normalized difference built-up index (NDBI) in 2001 and 2019 were retrieved using Landsat 7 Enhanced Thematic Mapper + (ETM +). Our results showed the following: (i) an increase in the mean temperature of 4.02 °C from 2001 to 2019 in MM, (ii) rise in NDBI and a decline in NDVI values is correlated with an increase in LST over the last years in MM; (iii) LULC maps affirmed the highest occupancy of built-up areas caused by possible urban sprawl explaining the increase in warming in MM. This study provides added evidence that urbanization activities in Metro Manila leads to a warming effect that drives discomfort to many average Filipinos. Hence, environmental policies that would implement proper urban design utilizing more green spaces is recommended for local government units in MM to support.”

Title: When the city heats up: mapping urban heat risks through environmental and socioeconomic factors in Quezon City, Philippines
Author: Aerol Cedrick Treyes

On the 31% coverage and overlap with high-risk areas: From the **Abstract**: "Spatial analysis revealed that 81% of barangays, accounting for 70% of the city's population, fall under high-risk HRI classification (0.61-0.80)... High-risk areas are concentrated in the southern and southwestern zones of the city, characterized by dense urbanization, limited vegetation, and high surface temperatures. **Intraurban heat islets, covering 31% of the study area, strongly overlap with high-risk HRI barangays, emphasizing the compounded effects of environmental and socioeconomic factors on heat risk.**"

On localized clustering and non-random thermal variation: While the provided text does not use the exact statistical terms "spatial autocorrelation" or "non-random," it explicitly confirms that the heat variation is spatially clustered into specific zones and hotspots.

"Within cities, however, **localized hotspots known as intraurban heat islets can emerge, where clusters of elevated temperatures collectively make up the broader UHI pattern [5]**." ... "These risks are not uniformly distributed across the city, as levels of urbanization, environmental conditions, and social vulnerability vary considerably between areas."

"Elevated LST values are indicative of intraurban heat islets, where localized temperature increases exacerbate heat risks [3, 24]."

Title: Urban Effects on Land Surface Temperature in Davao City, Philippines
Author: Myla Ross Tinoy, A. U. Novero, K.P. C. Landicho, Alvin Baloloy, Ariel Blanco

"The relationships between the factors and LST were studied through multiple and quantile regression models (MRM & QRM). Results showed that variable PM10 does not have any significance in the MRM. Meanwhile, QRM were fitted to different quantile values, namely: 10th, 25th, 50th, 75th, and 90th. It is only at the 90th quantile where all the independent variables are good predictors for the LST. Albedo is the most important predictor for the LST at 10th quantile whereas Elev for the 25th quantile. However, when LST is at the 50th, 75th, and 90th quantiles NDBI is the most significant variable at predicting LST."

"Moreover, at the Southeast of Davao, HS were dominated by the main urban center (whole coverage of Districts of Poblacion and Agdao; and few areas of Districts of Talomo and Buhangin) and coastal barangays of Bunawan and Toril Districts."

"The regression regular coefficient estimates embody the mean change in the response variable given a one unit of change in the predictor variable while holding other predictors in the model constant. Interpreting the model is as follows: 0.01 unit increase in EVI decreases LST by 9.064°C; 0.01 unit increase in NDBI increases LST by 0.06235°C, and; 0.01 unit increase in Albedo decreases LST by 0.5369°C."

Title: Spatiotemporal Analysis Of Urban Land Surface Temperature (Ulst) In Davao City
Author: Marvin Batican and Jason Ben Paragamac

"This study investigated the spatiotemporal dynamics of urban land surface temperature (ULST) from 2015 to 2024 across the city's three congressional districts. Using Landsat 9 satellite imagery and geospatial processing in QGIS, annual land surface temperature (LST) trends were derived and analyzed. Results revealed a consistent increase in both minimum and maximum LST values across all districts, with 2024 recording peak temperatures nearing or exceeding $38^{\circ}C$. The most pronounced warming occurred in District I, particularly in

highly urbanized barangays such as 1-A, 7-A, and 11-B."

"By analyzing the areas with constant increase in land surface temperature, the data revealed that all of the top 10 urban barangays with the highest land surface temperature (LST) from 2015 to 2024 are located within the 1st Congressional District of Davao City. Barangay 1-A ranks highest, followed by Barangays 7-A, 11-B, and 2-A. The concentration of high-LST barangays within a single district suggests a notable spatial pattern of urban heat distribution, which may be attributed to higher levels of built-up areas, reduced vegetation cover, dense population, and intensive commercial and industrial activities commonly found in central urban zones."

Title: GIS-Based Spatio-Temporal Analysis of Vegetation Dynamics, Urban Expansion, and Land Surface Temperature in Davao City [PAYWALLED] [ABSTRACT]
Author: Richelyn C. Dumdum Jason Ben R. Paragamac

“The Earth's surface and atmosphere have been substantially modified by the growth of anthropogenic activities and urbanization. Changes in the composition and operation of the urban ecological system will affect the health of city dwellers by increasing the risk of heat exposure, increasing the risk of violence, illness, and mortality, and reducing labor productivity. Davao City is currently undergoing rapid urbanization, and no comprehensive study has examined the effects of the changes on vegetation indices, urban expansion, traffic density, and land surface temperature, which are likely to affect the area. This study analyzes the spatio-temporal change in vegetation indices, built-up expansion, traffic density, and land surface temperature from 2018 to 2022 using Landsat 8 imagery and GIS tools, particularly QGIS. The results show a moderate negative relationship between NDBI and NDVI ($r = -0.4313$), a strong positive relationship between NDBI and LST ($r = 0.6216$), and a weak negative relationship between NDVI and LST ($r = -0.2495$). Results highlight the interconnected influence of urbanization, vegetation dynamics, and thermal conditions within the study area, underscoring the need to integrate green infrastructure, particularly along major road networks, to mitigate rising land surface temperatures.”

Title: Assessment of Intra-Urban Heat Island in a Densely Populated City Using Remote Sensing: A Case Study for Manila City
Author: Mark Angelo Purio,Tetsunobu Yoshitake, and Mengu Cho

"To address this, we present the use of satellite-derived (2013–2022) and meteorological data (2014–2020) to assess intra-urban heat islands in Manila City, Philippines. The assessment includes (a) understanding the temporal variability of air temperature measurements and outdoor thermal comfort based on meteorological data, (b) comparative and correlative analysis between common Land-Use Land Cover indicators (Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI) and Normalized Difference Built-up Index (NDBI)) and Land Surface Temperature (LST), (c) spatial and temporal analysis of LST using spatial statistics techniques, and (d) generation of an intra-urban heat island (IUHI) map with a recommended class of action using a suitability analysis model."

"In particular, densely packed structures with little greenery develop "islands" with greater temperatures than their surroundings [9,11–13]."

Title: Thermal changes along the urban-rural continuums in Southeast Asia
Author: Yuanhui Zhu*, Soe W Myint, Jiquan Chen, Peilei Fan, Karen C Seto, Atul Jain, Jiaguo Qi and Jin Wang

"This study examines LST, LCLU, and NDVI changes across 19 SEA cities of varying urbanization levels from 2000 to 2022 using Landsat images. Results show that LST change rates differ by urbanization levels, with built-up lands driving urban warming. Larger cities exhibited more intense and widespread LST increases extending into rural areas. The megacities experienced significant LST growth within 5 km of city centers despite limited built-up expansion. However, some areas in many cities experience decreasing trends of heat. NDVI significantly influenced LST changes through its cooling effect and the warming impact due to vegetation loss caused by urban expansion."

Discussing the 19 cities and the correlation with built-up land and NDVI (vegetation):
"We summarized the LST Sen's slope across 19 cities along the urban-rural gradient (figure 4). For the entire study area (figure 4(a)), the LST Sen's slope peaked at the city center (0.0303 °C/year), decreased to 3 km, and followed a bimodal pattern with peaks at 6 km and 17 km before declining. NDVI Sen's slope increased consistently from urban to rural areas, while the built-up Sen's slope mirrored the LST trend, and cropland exhibited an opposite pattern. Significant correlations were observed between the LST Sen's slope and built-up (positive, R^2 = 0.136), cropland, wetland, and NDVI (negative, R^2 = 0.192, 0.150, and 0.355, respectively)."

Detailing the specific extreme case of the megacity (Jakarta), confirming the intense NDVI/vegetation loss driving heat:

"In the megacity (figure 4(b)), the LST Sen's slope exhibited a trimodal pattern... While the city center showed an LST Sen's slope close to 0, the rest of the megacity experienced an overall upward LST trend, indicating rapid urban warming over the past two decades. Interestingly, the NDVI Sen's slope displayed a rapidly decreasing trend from the city center, with valleys at 16 km, 20 km, and 25 km, contrasting the LST Sen's slope pattern. Built-up areas, the dominant LCLU type, initially showed an opposite trend to LST within the first 5 km but mirrored the LST pattern beyond this distance... Regression analysis revealed significant negative correlations between the LST Sen's slope and cropland (R^2=0.359) vegetation (R^2=0.153), and NDVI (R^2=0.792)."

Title: Integrating Satellite-Based and Participatory Spatial Analyses for Local, Contextualised, and Nuanced Extreme Heat Hotspot Mapping in Overheated Urban Southeast Asia
Author: Laurence L. Delina, Homer Pagkalinawan, Sharon Feliza Ann Macagba

"In addition, higher population density, dense road networks, and proximity to extensive heat-retaining infrastructure (notably airports) are consistently associated with elevated temperatures, whereas greater greenspace accessibility is associated with cooling."

"Notably, city-level mechanisms differ." "Metro Manila's model identifies strong contributions from primary roads (coefficient = 0.540, p<0.01) and proximity to the airport (1.185, p<0.01) and displays a large, significant spatial lag parameter (≈0.916 p<0.01), indicating that heat in one grid cell is strongly influenced by neighbouring cells—evidence of spatial spillover at neighbourhood scales."

"Bangkok's model, by contrast, highlights the cooling effect of greenspace accessibility (green accessibility ratio ≈-0.00043, p<0.01) and the warming influence of trunk roads (0.181, p<0.01), while its very large spatial error coefficient (≈0.980, p<0.01 suggests much of the remaining temperature variation is spatially structured and tied to omitted local factors (microclimate heterogeneity, fine scale morphology, or unmeasured materials)."

"In Metro Manila, extreme heat clusters closely follow north-south transport arteries—most notably along Epifanio delos Santos Avenue (EDSA) and other principal arterials." "The regression coefficient for primary roads (0.540***) is large and highly significant, indicating a strong positive association between road-dominated grid cells and LST after accounting for built form, green access, and socioeconomic factors." "This pattern points to the combined thermal effects of asphalt surfaces, high traffic density, and associated anthropogenic heat emissions concentrated along these corridors."
"By contrast, Bangkok exhibits a contiguous central hot zone spanning 33 of its 50 districts, with heat propagation aligned with trunk roads that form linear corridors radiating from the core." "The road coefficient here (0.181) is positive and significant but smaller than Manila's, suggesting that while major roadways contribute to warming, Bangkok's heat field is also shaped by a tightly clustered set of local factors and urban morphology that create an extended central heat nucleus rather than isolated linear hotspots."

"Together these methods reveal two policy-relevant insights that are particularly important for urban heat action in resource-constrained tropical cities: (1) a consistent, nonlinear (quadratic) relationship between a building efficiency ratio and local temperatures... and (2) the strong, systematic roles of population density, dense road networks, and large heat-retaining infrastructure (notably airports) in elevating local temperatures, while greenspace accessibility is associated with cooling."